

UNIVERSITY OF BATH • AUGUST 2026

Formalization and physics: a new frontier for AI



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[leanprover-community/physlib](https://leanprover-community.github.io/physlib)



josephtoobysmith.com

A new era of proving

MAY 2026

The unit-distance problem

Erdős, 1946.
Disproved.

OpenAI, internal model

JUL 2026

The Jacobian conjecture

Keller, 1939.
Counterexample in $\mathbb{C}^3$.

Alpoge, with Claude Fable 5

AUG 2026

Ten open problems

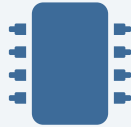
Each with a
Lean 4 certificate.

OpenAI “Astra”

Two ways of verifying



Expert review



Computer

A rapid overview of Lean

One way to check by computer

Definitions, theorems, proofs



consistency & correctness, guaranteed

Built on type theory

Lean knows the **type** of everything, and checks that things have the type expected.

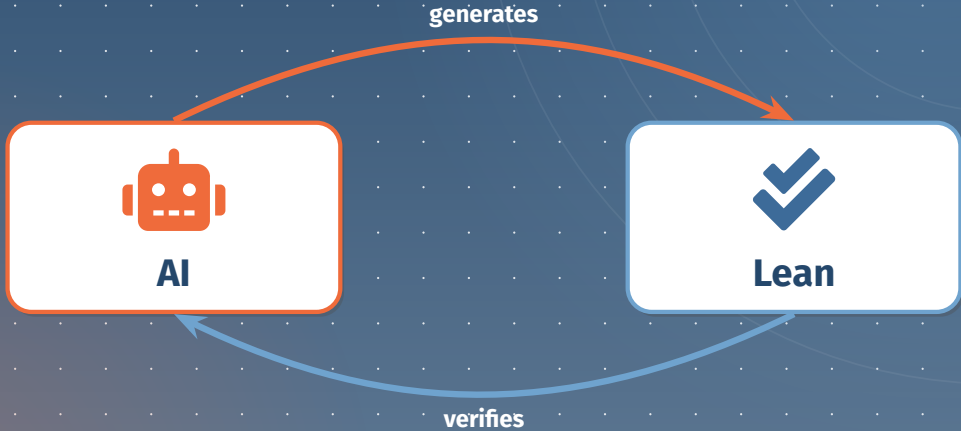
$$\underbrace{p}_{\text{proof}} : \underbrace{P}_{\text{theorem}} \quad \underbrace{7}_{\text{term}} : \underbrace{\mathbb{N}}_{\text{type}}$$

A definition, a theorem, a proof

```
def energy (m v : Real) : Real :=  
  m * v ^ 2 / 2
```

```
theorem energy_nonneg  
  (m v : Real) (hm : 0 <= m) :  
  0 <= energy m v := by  
  unfold energy  
  positivity
```

A feedback loop



What the loop is built on



leanprover-community / mathlib4

A *monolithic* library of mathematics formalized in Lean — large enough to work at a high level.

public

Lean 4

● Lean 📄 100,000 definitions ✓ 200,000 theorems 👤 500+ contributors ⌚ started 2017

My broad interest



leanprover-community / physlib

A community-driven project to digitalise results from physics into Lean.

public

Lean 4

● Lean ★ 676 stars 🔗 153 forks 👤 85 contributors ⌚ started 2024

TWO THINGS

What I am interested in now



Local models, and citizen science

THE QUESTION

What can we do *now*, with the models people already have?

• Docstring for `rep` incorrectly refers to Lorentz vectors instead of covectors #1495

open

good first issue

opened by jstoobysmith

- *-- The representation of the Lorentz group on Lorentz vectors. --*

+ *-- The representation of the Lorentz group on Lorentz covectors. --*

Found with a local LLM pass over Physlib docs (*qwen3.5:9b*); 10 of 10 independent verification runs agreed before filing.

Formalizing the Standard Model

$$\begin{aligned}\mathcal{L}_{\text{SM}} = & -\frac{1}{4} G_{\mu\nu}^A G^{A\mu\nu} - \frac{1}{4} W_{\mu\nu}^I W^{I\mu\nu} - \frac{1}{4} B_{\mu\nu} B^{\mu\nu} + \sum_{\psi} i \bar{\psi} \gamma^{\mu} D_{\mu} \psi \\ & + (D_{\mu} H)^{\dagger} (D^{\mu} H) + \mu^2 H^{\dagger} H - \lambda (H^{\dagger} H)^2 \\ & - \left(\bar{Q}_L Y_u \tilde{H} u_R + \bar{Q}_L Y_d H d_R + \bar{L}_L Y_e H e_R + \text{h.c.} \right)\end{aligned}$$

$$\begin{aligned}D_{\mu} &= \partial_{\mu} - ig_3 G_{\mu}^A T^A - ig_2 W_{\mu}^I t^I - ig_1 Y B_{\mu}, & B_{\mu\nu} &= \partial_{\mu} B_{\nu} - \partial_{\nu} B_{\mu}, \\ G_{\mu\nu}^A &= \partial_{\mu} G_{\nu}^A - \partial_{\nu} G_{\mu}^A + g_3 f^{ABC} G_{\mu}^B G_{\nu}^C, & W_{\mu\nu}^I &= \partial_{\mu} W_{\nu}^I - \partial_{\nu} W_{\mu}^I + g_2 \epsilon^{IJK} W_{\mu}^J W_{\nu}^K.\end{aligned}$$

$G = (SU(3) \times SU(2) \times U(1))/\Gamma$, acting on three generations of $Q_L(\mathbf{3}, \mathbf{2})_{1/6}$, $u_R(\mathbf{3}, \mathbf{1})_{2/3}$, $d_R(\mathbf{3}, \mathbf{1})_{-1/3}$, $L_L(\mathbf{1}, \mathbf{2})_{-1/2}$, $e_R(\mathbf{1}, \mathbf{1})_{-1}$, and $H(\mathbf{1}, \mathbf{2})_{1/2}$.

Thank you for watching



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